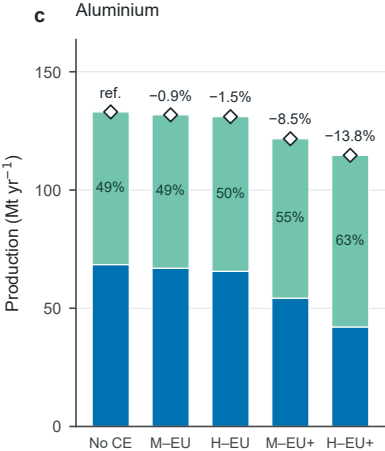
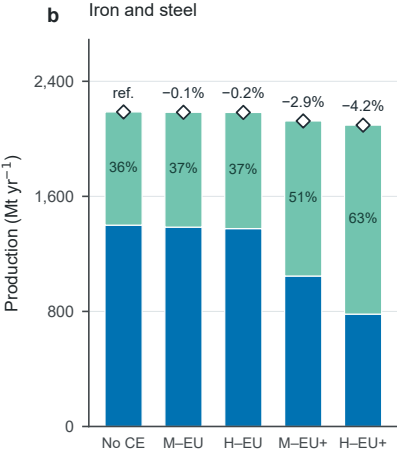
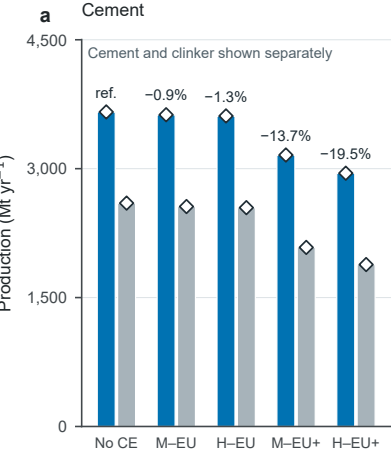


# Circular economy changes material demand and primary supply

Global production in 2050; all ten scenarios

Cement / primary metal    Clinker    Secondary metal    ◇ NDC output



Bars: Baseline. Diamonds: NDC. Labels above bars: Baseline change from no CE.

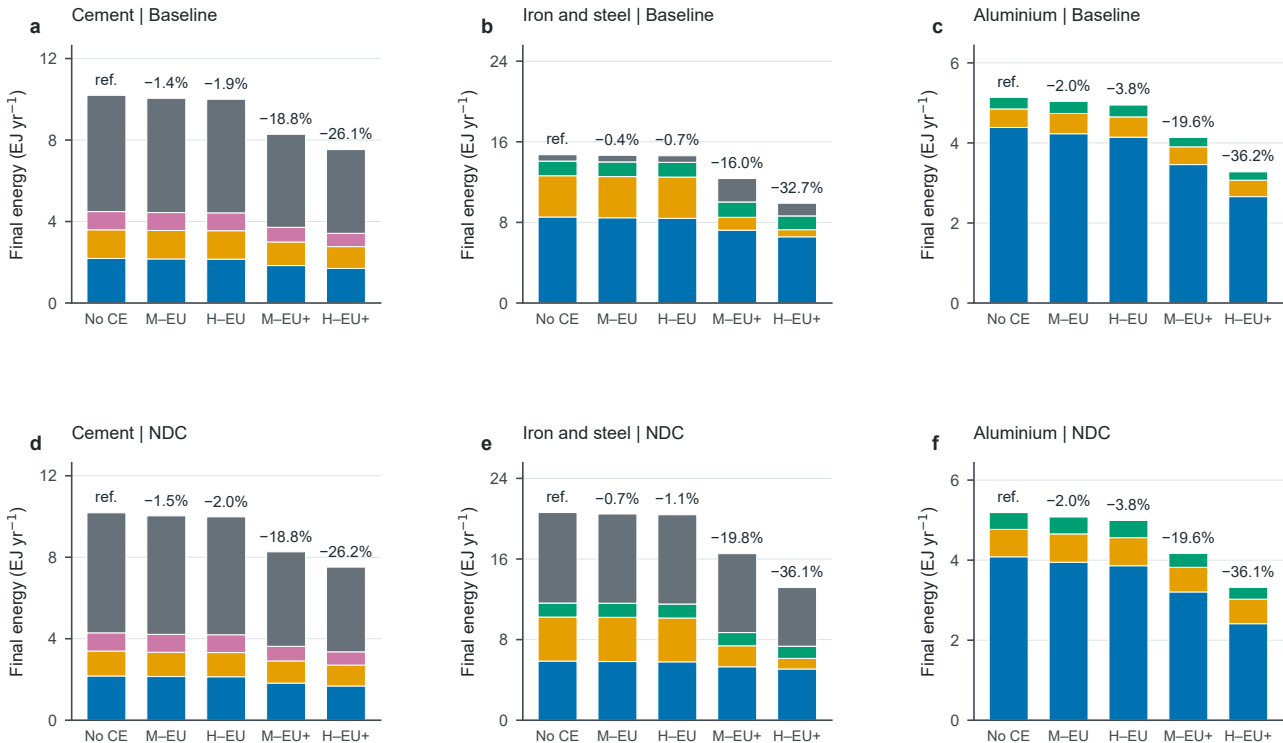
Labels within metal stacks: share of secondary production.

M/H = medium/high CE; EU+ = EU + international partners (17 model regions).  
Draft assumption: missing additive activity = zero. Baseline and NDC are model climate pathways.

Circular economy reduces final energy demand and changes the fuel mix

Global final energy in 2050; Baseline and NDC

Electricity   Gases   Hydrogen   Liquids   Solids



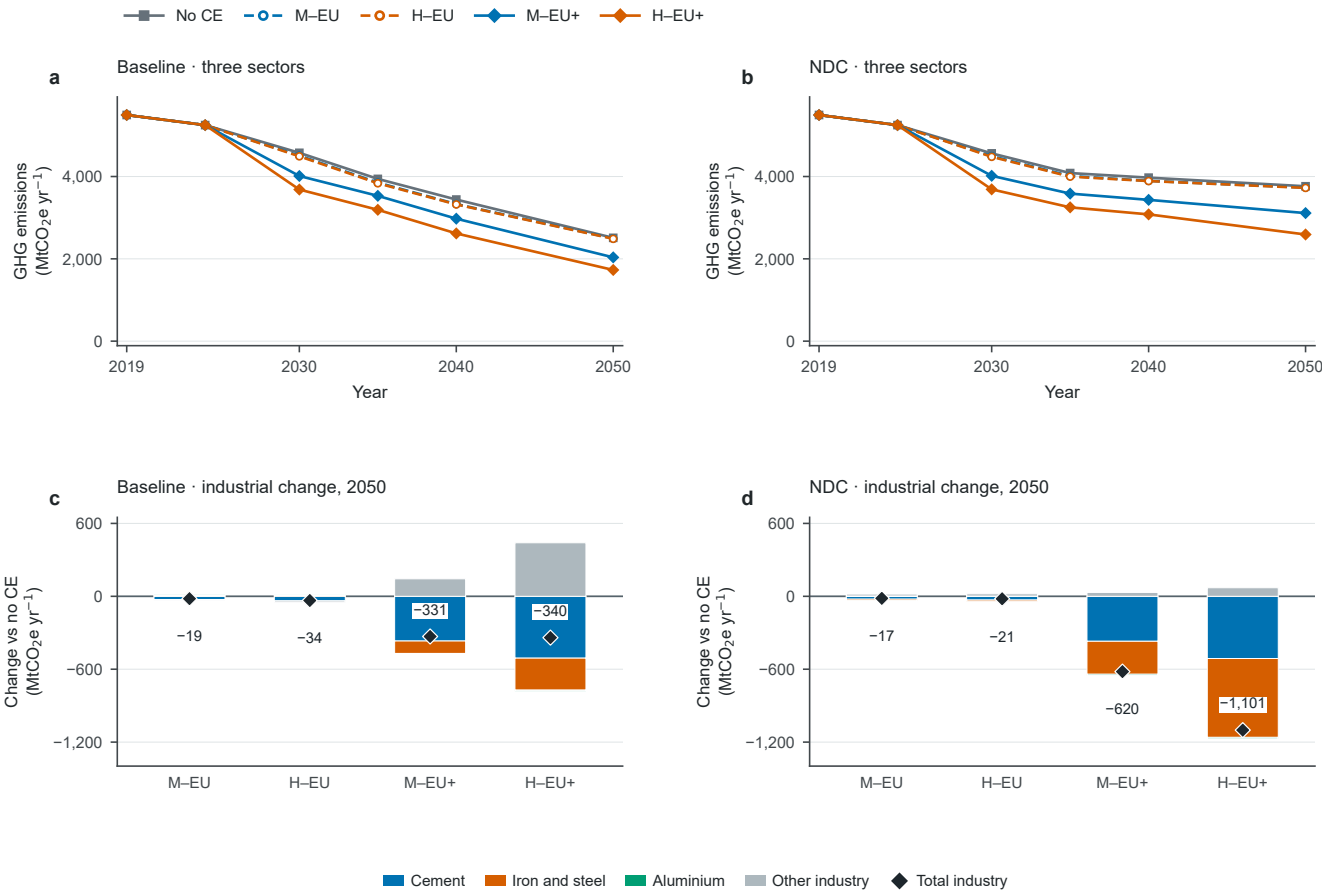
Labels: change from no CE under the same climate pathway. Sector scales are identical across rows.

M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Drافت assumption: missing additive activity = zero. Baseline and NDC are model climate pathways.

# CE emissions savings and the wider industrial response

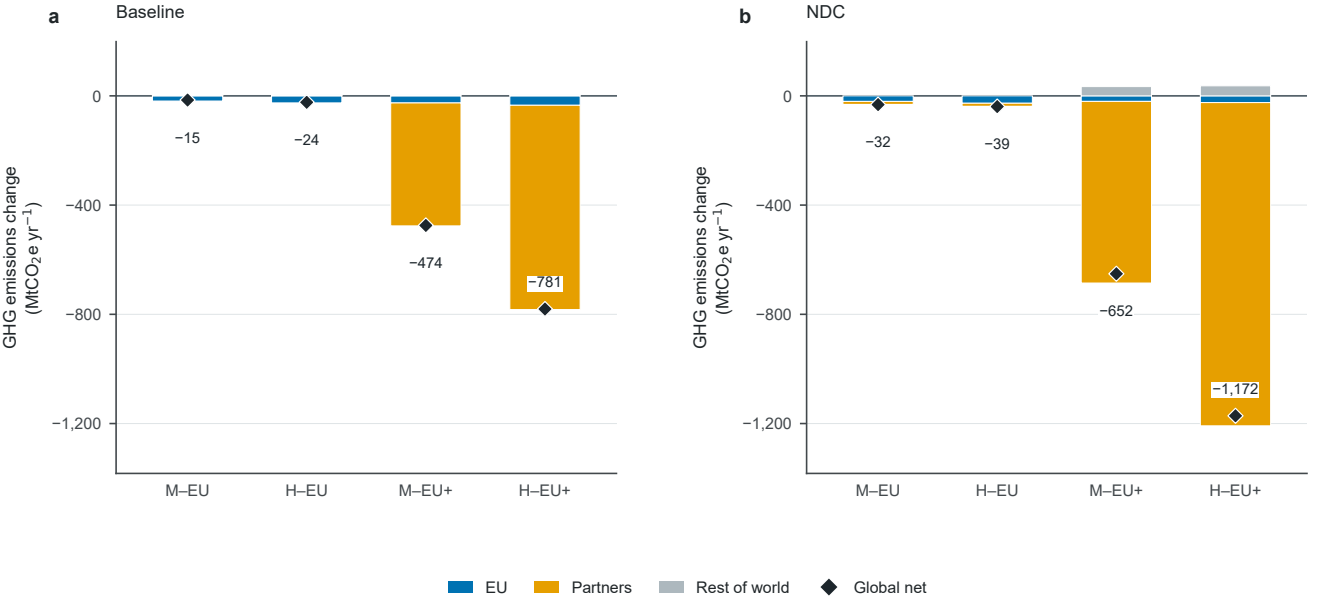
Global greenhouse-gas emissions · trajectories and matched 2050 changes



M/H = medium/high CE; EU+ = EU + international partners (17 model regions).  
Draft assumption: missing additive activity = zero. Baseline and NDC are model climate pathways.

# Where CE emissions benefits occur, and what partners add

Three sectors · signed regional changes vs matching no CE · 2050

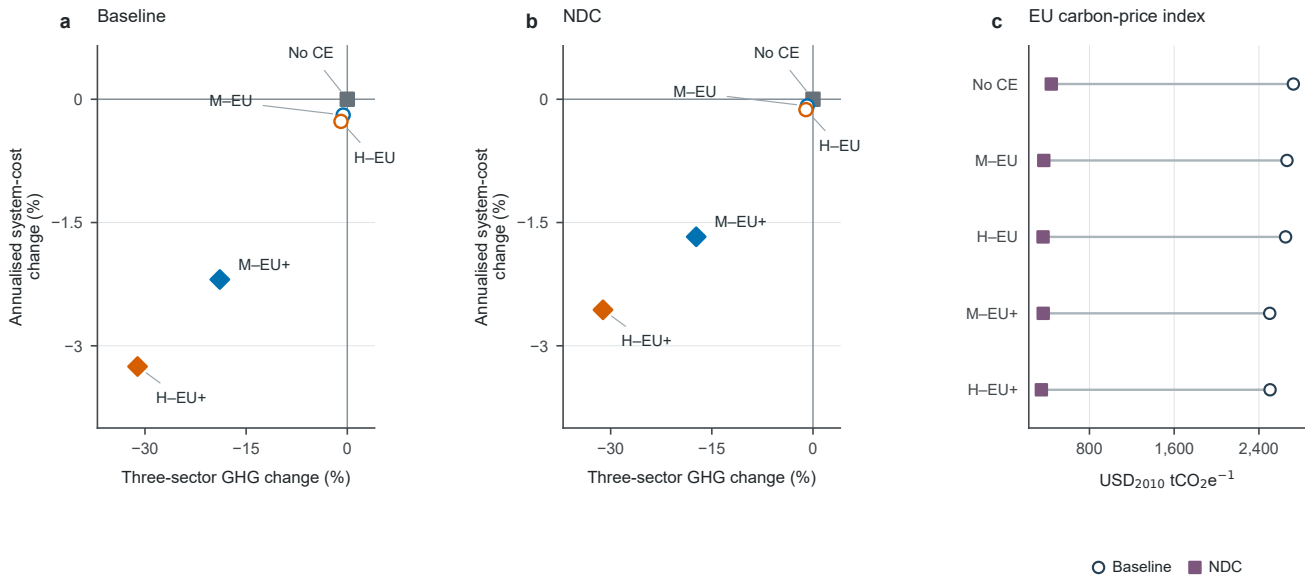


M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Draft assumption: missing additive activity = zero. Baseline and NDC are model climate pathways.

## CE, annualised system costs and EU carbon-price requirements

Changes relative to climate-matched no CE · 2050



M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Draft assumption: missing additive activity = zero. Baseline and NDC are model climate pathways.